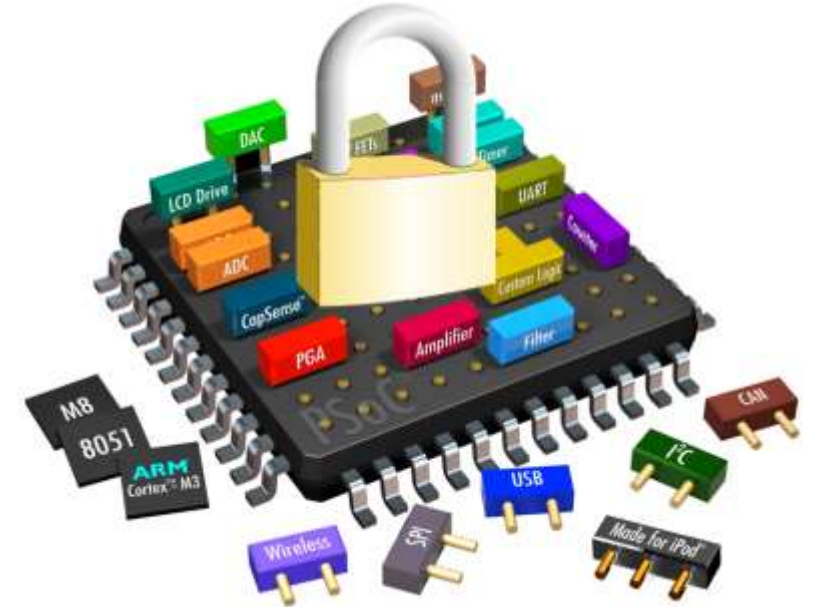


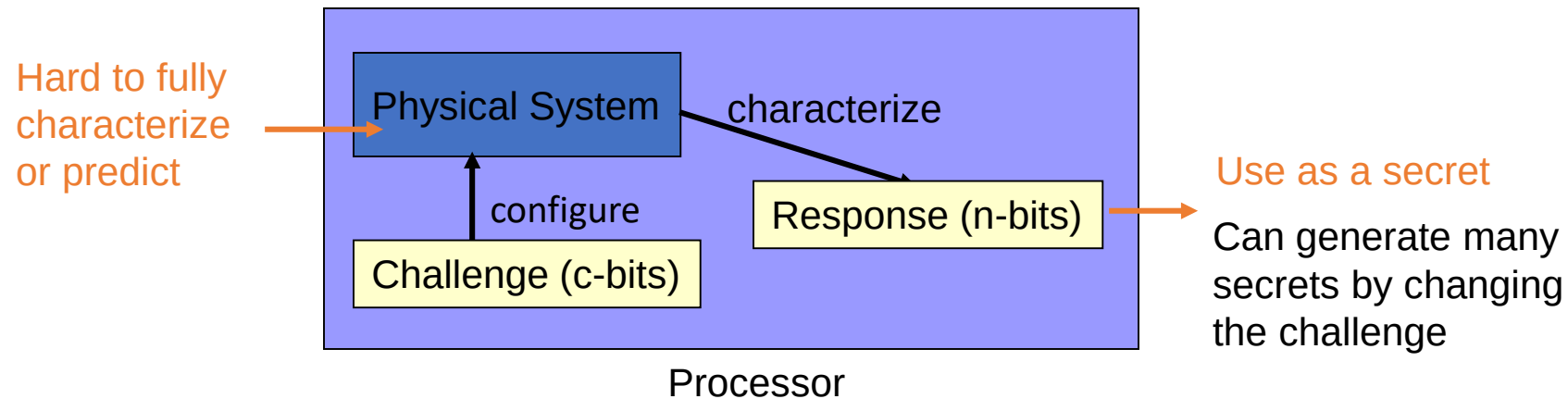
Hardware Security

Physical Unclonable Function (PUF)



Physical Unclonable Functions (PUFs)

- Generate keys from a **complex physical system**
- Security advantage
 - Keys are generated on demand → No non-volatile secrets
 - No need to program the secret
 - Can generate multiple master keys
- What can be hard to predict, but easy to measure?

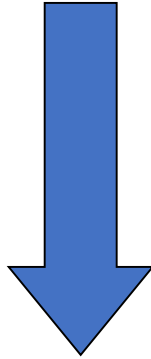


Definition

- A Physical Random Function or Physical Unclonable Function (PUF) is a function that is:
 - Based on a physical system
 - Easy to evaluate (using the physical system)
 - Its output looks like a random function
 - Unpredictable even for an attacker with physical access

WYSINWYG

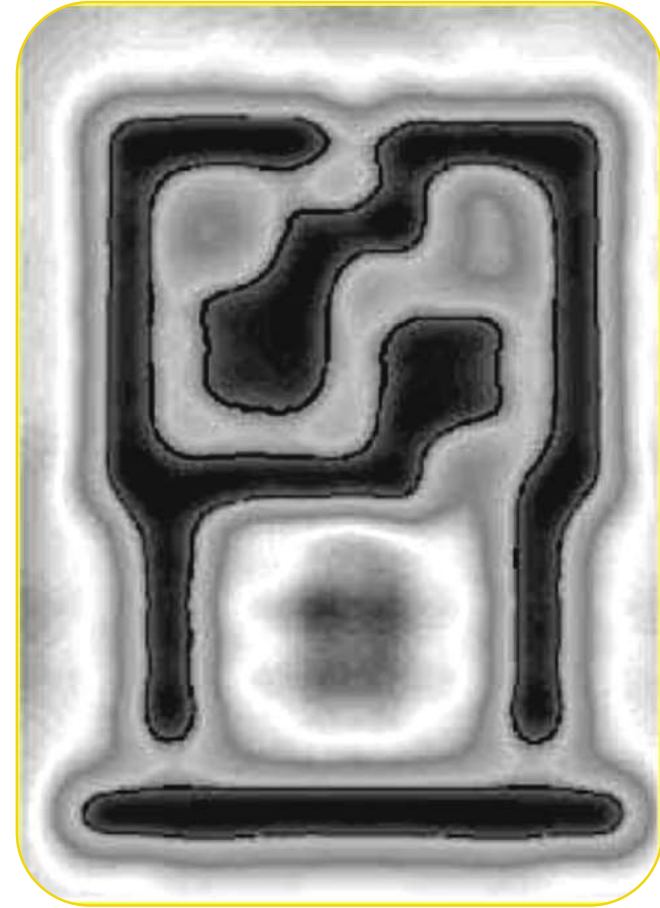
Sub-Wavelength **WYSINWYG**



What You See Is Not What You Get

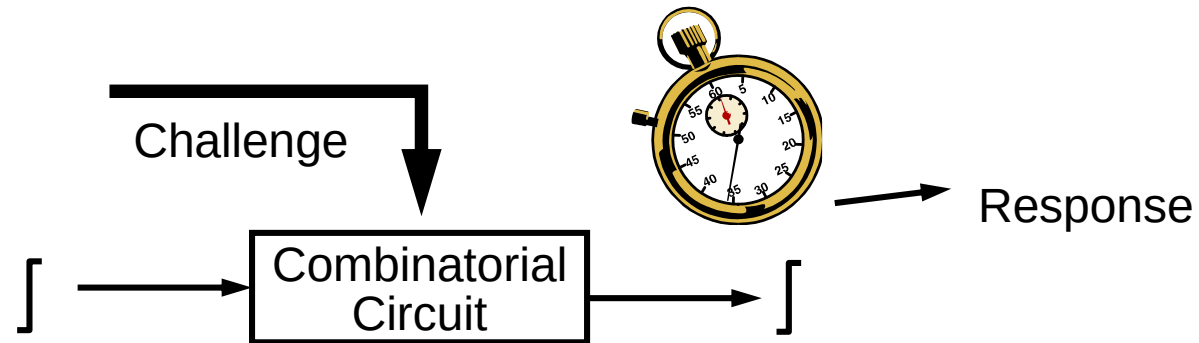
Process variations

No two transistors have the same parameters



Example: Delay PUF

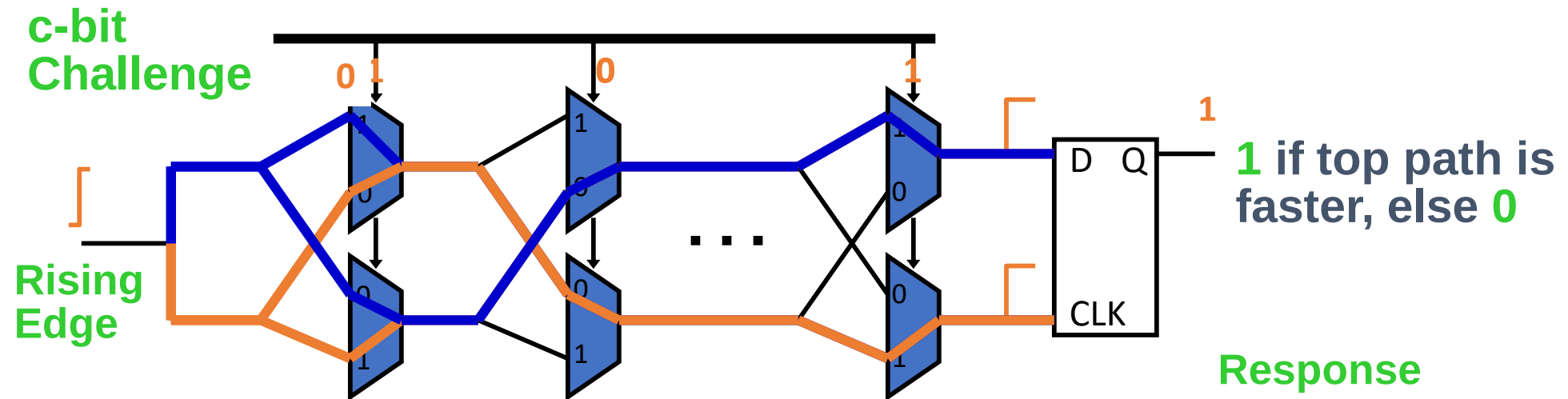
- Because of process variations, no two integrated circuits are identical
- Experiments in which identical circuits with identical layouts were placed on different FPGAs show that path delays vary enough across ICs to use them for identification.



Physical Attacks

- Make PUF delays depend on overlaid metal layers and package
- Invasive attack (e.g., package removal) changes PUF delays and destroys PUF
- Non-invasive attacks are still possible
 - To find wire delays one needs to find precise relative timing of transient signals as opposed to looking for 0's and 1's
 - Wire delay is not a number but a function of challenge bits and adjacent wire voltages and capacitances

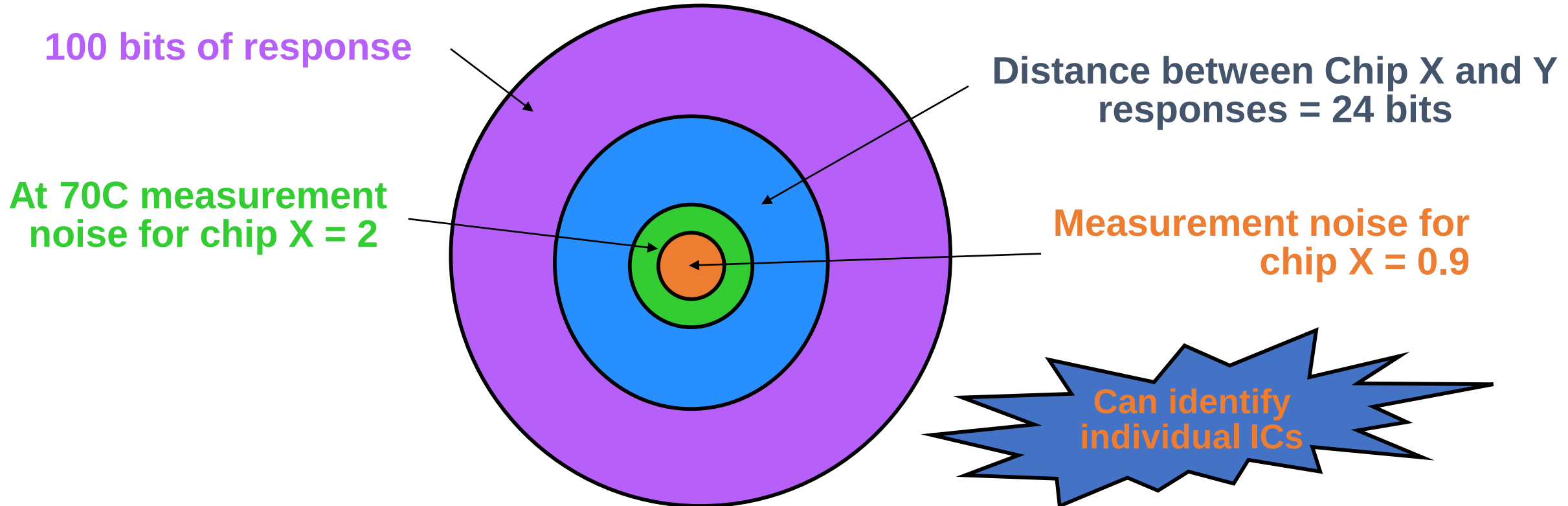
Arbiter PUF



- Compare two paths with an identical delay in design
 - Random process variation determines which path is faster
 - An arbiter outputs 1-bit digital response
- Path delays in an IC are statistically distributed due to random manufacturing variations

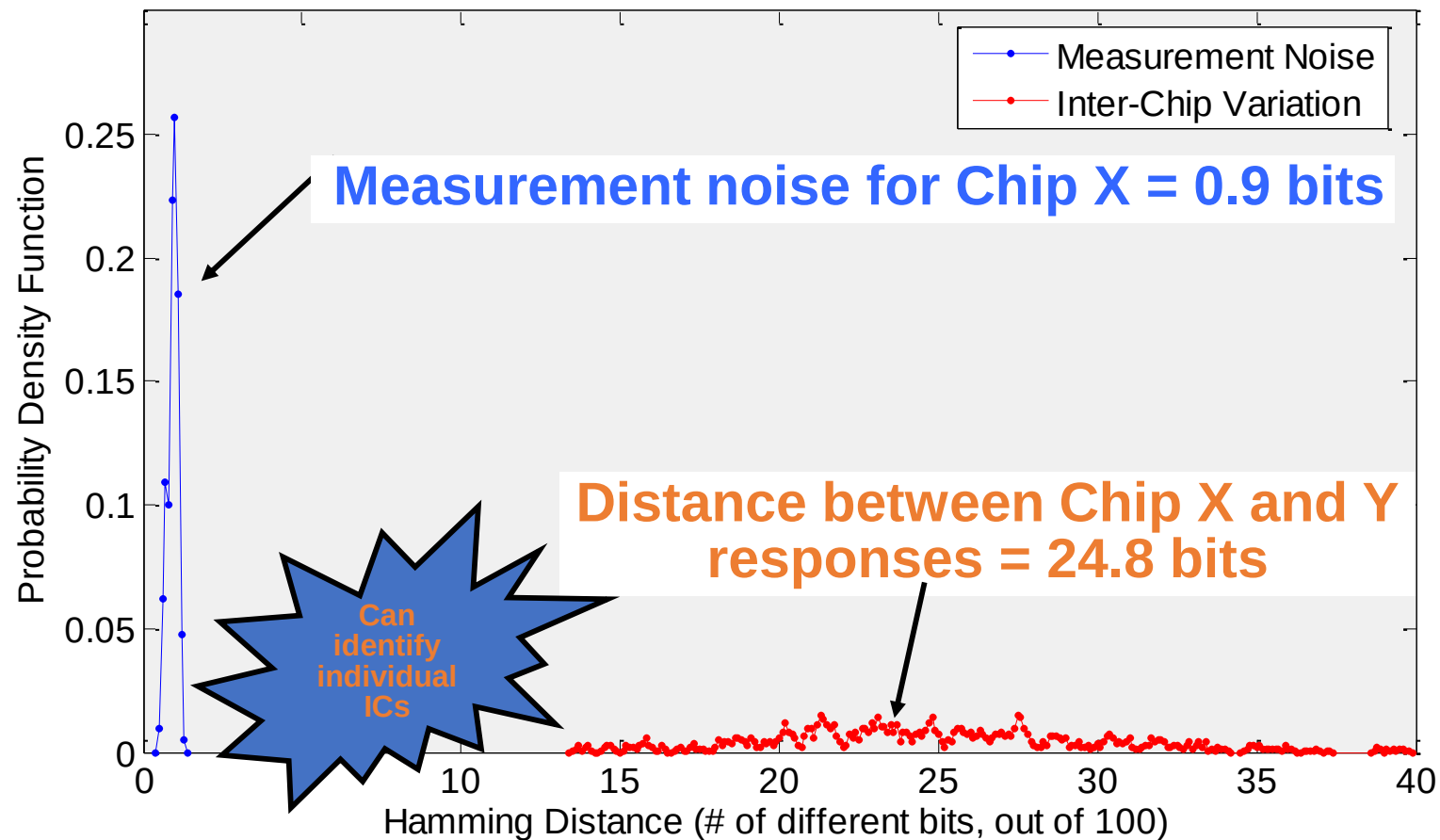
Experiments

- Fabricated candidate PUF on multiple ICs, 0.18um TSMC
- Apply 100 random challenges and observe responses



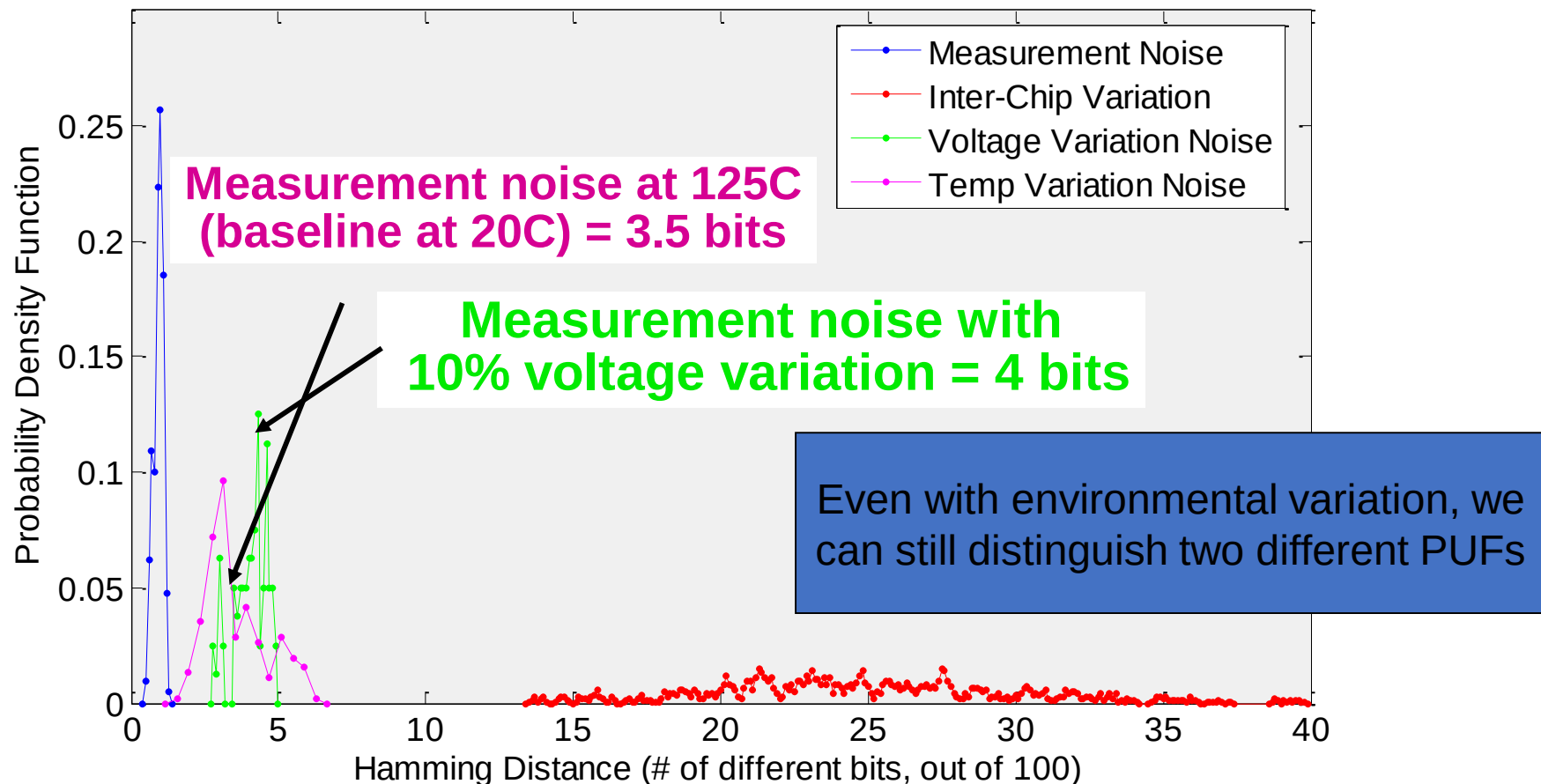
Inter-Chip Variation

- Apply random challenges and observe 100 response bits

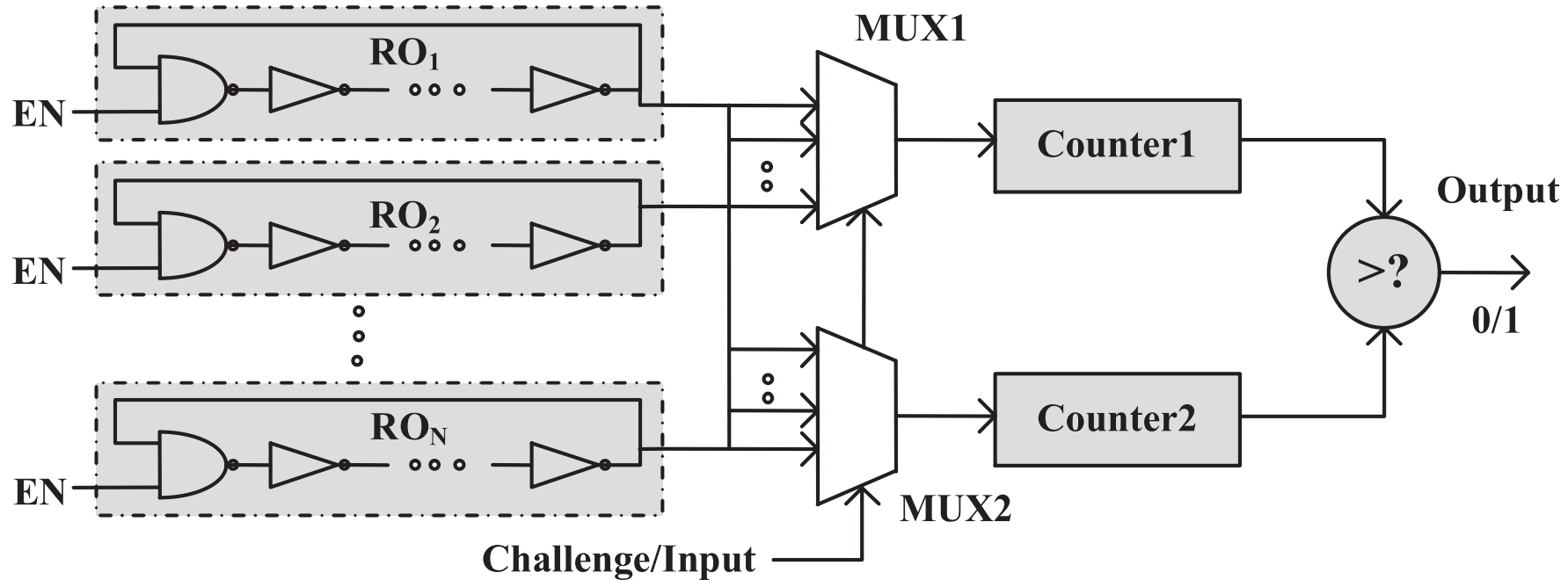


Environmental Variations

- What happens if we change voltage and temperature?



RO PUF



- The structure relies on delay loops and counters instead of MUX and arbiters
- Better results on FPGA – more stable

RO PUF Advantages

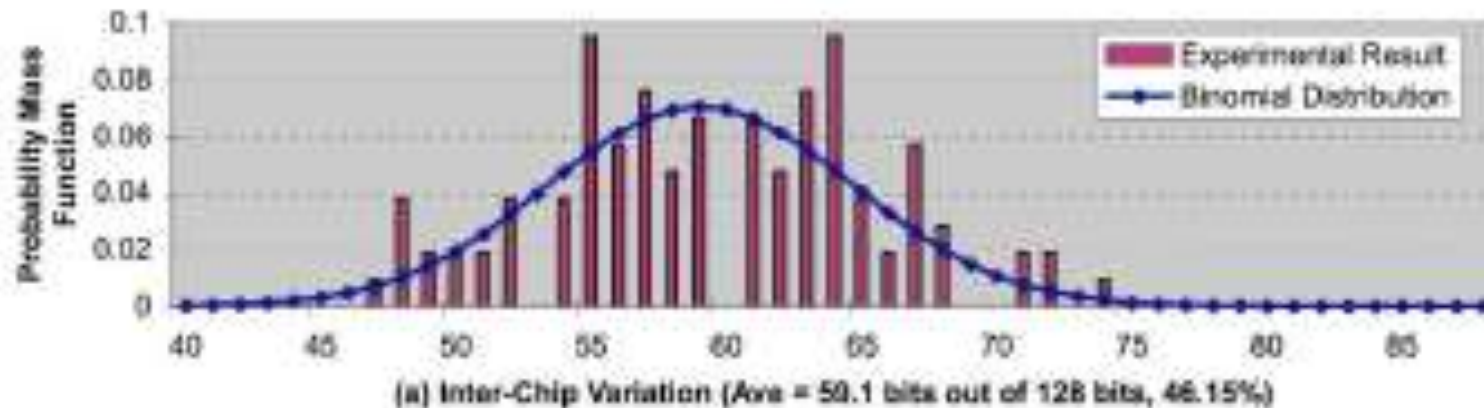
- ROs, whose frequencies are far, are more stable than the ones with closer frequencies
 - Possible advantage: do not use all pairs, but only the stable ones
 - It is easy to watch the distance in the counter and pick the very different ones.
 - Can be done during enrollment
- RO PUF allows an easier implementation for both ASICs and FPGAs
- The Arbiter PUF is appropriate for resource constrained platforms such as RFIDs and the RO PUF is better for use in FPGAs and in secure processor design.

Experiments with RO PUFs

- Experiments done on 15 Xilinx Virtex4 LX25 FPGA (90nm)
- They placed 1024 ROs in each FPGA as a 16-by-64 array
- Each RO consisted of 5 INVs and 1 AND, implemented using look-up tables
- The goal is to know if the PUF outputs are unique (for security) and reproducible (for reliability and security)

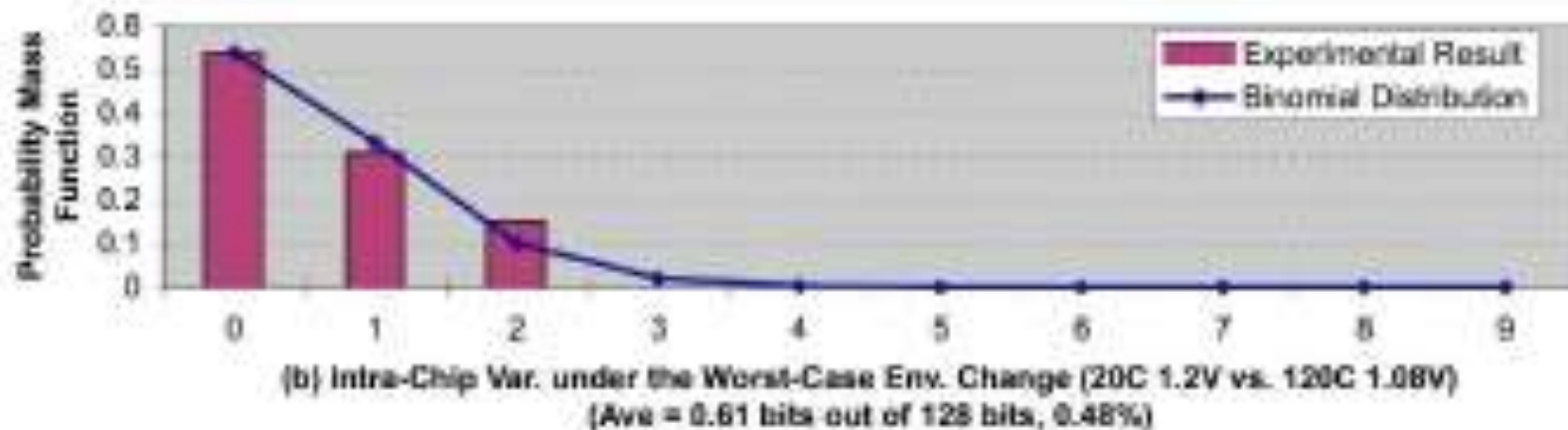
Inter-chip Variations

- 128 bits are produced from each PUF
- x-axis: number of PUF o/p bits different b/w two FPGAs; y-axis: probability
- Purple bars show the results from 105 pair-wise comparisons
- Blue lines show a binomial distribution with fitted parameters ($n=128$, $p=0.4615$)
- Average inter-chip variations 46.15%

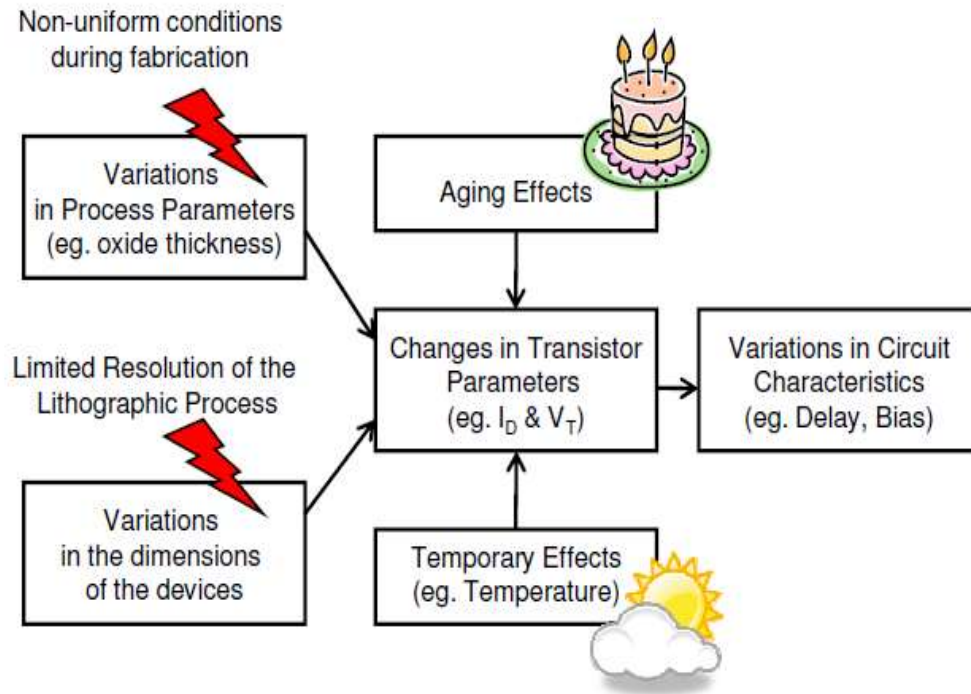


Intra-chip Variations

- PUF responses are generated at two different conditions and compared
- Changing the temperature from 20°C to 120°C and the core voltage from 1.2 to 1.08 altered the PUF o/p by ~0.6 bits (0.47%)
- Intra-chip variations is much lower than inter-chip – the PUF o/p did not change from small to moderate



Reliability of RO PUFs



- Two types of reliability issues:

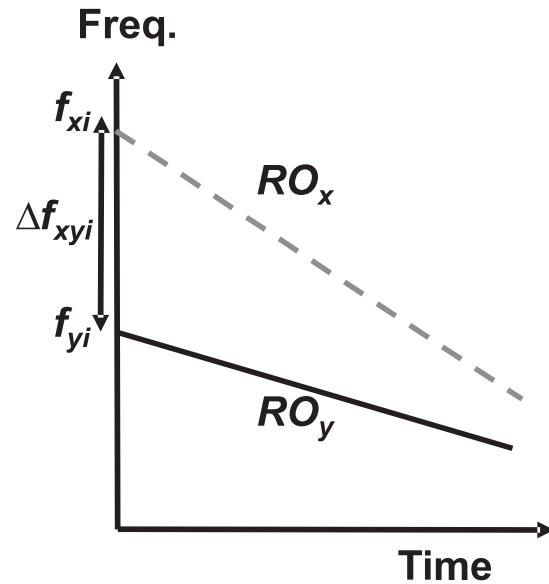
- Aging:

- Negative Bias Temperature Instability
- Hot Carrier Injection (HCI)
- Temp Dependent Dielectric Breakdown
- Interconnect Failure

- Temperature

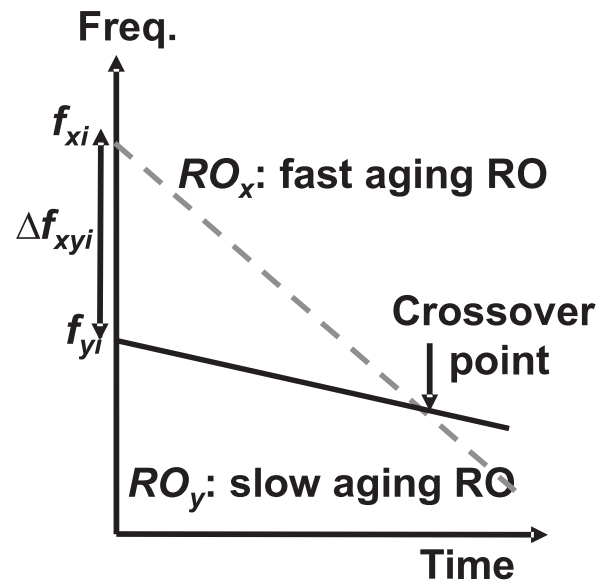
- Slows down the device

Bitflip due to Frequency Degradation



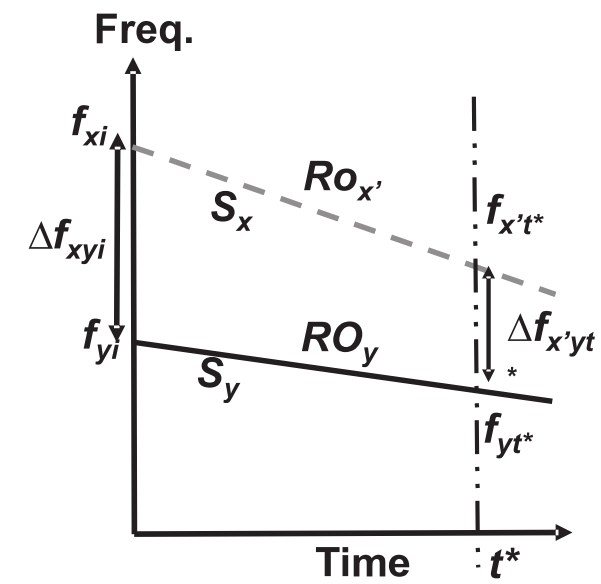
(A)

Moderate



(B)

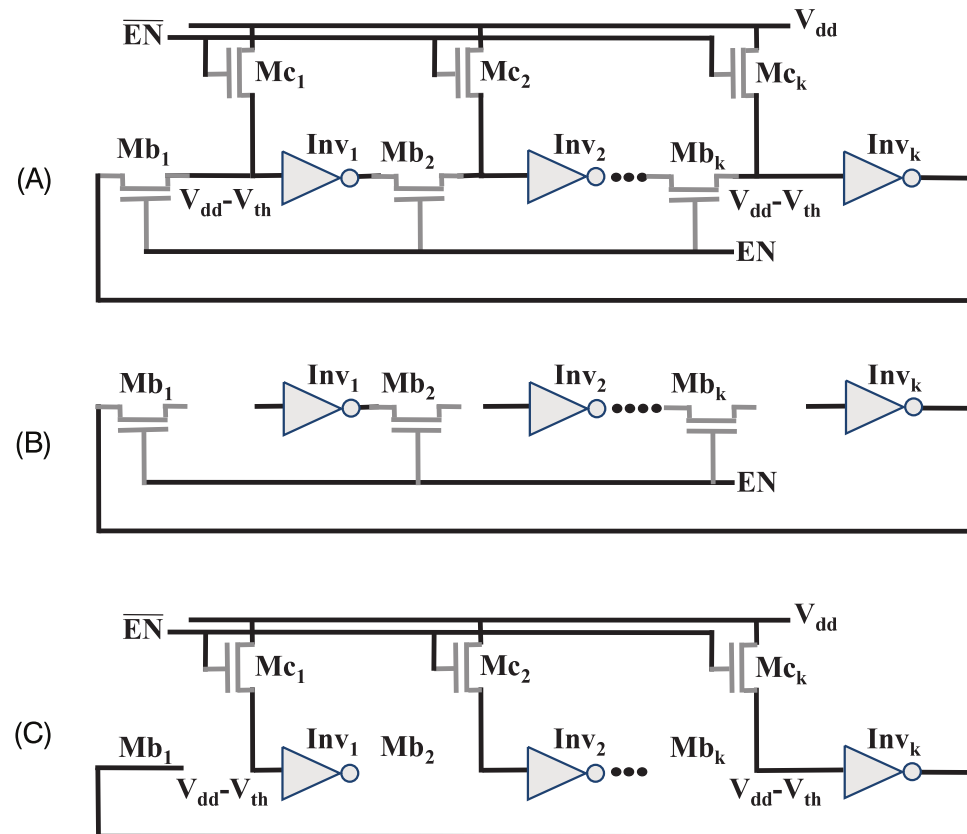
High degradation



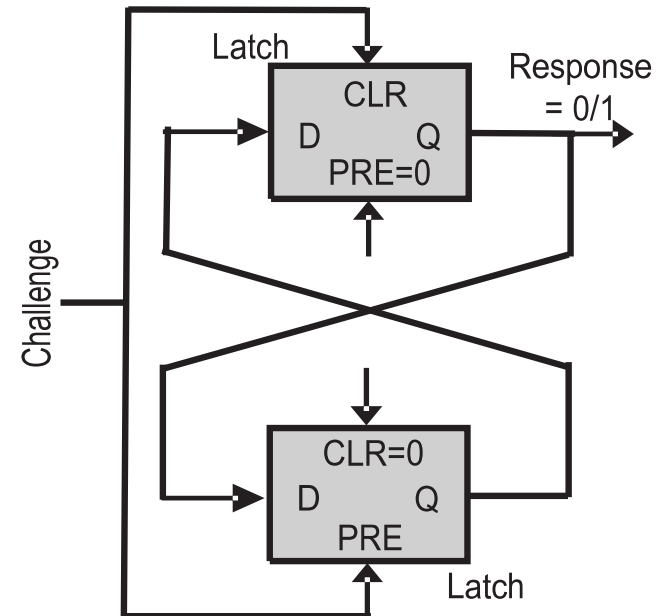
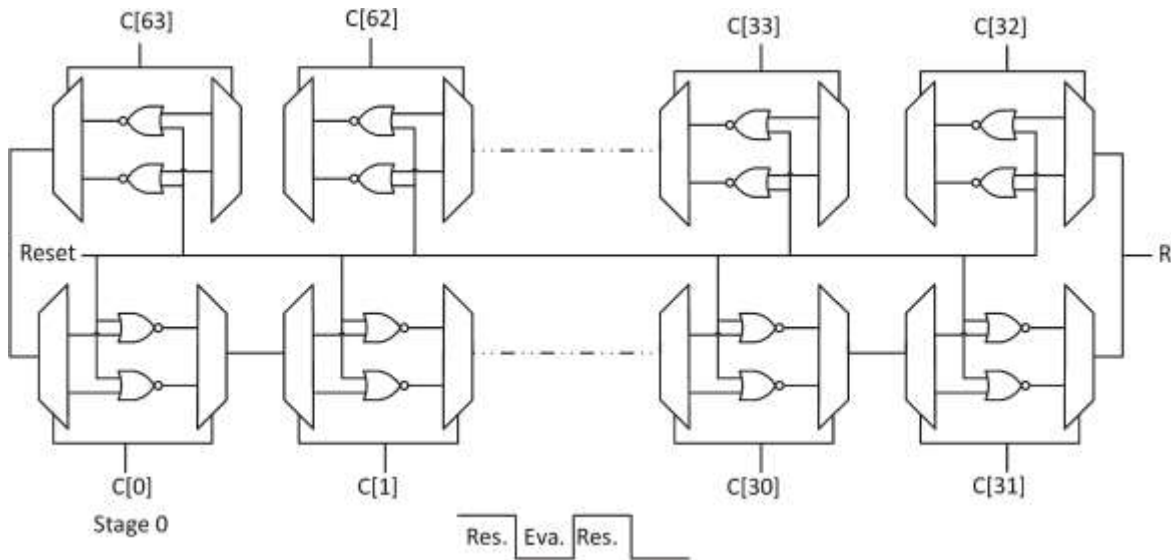
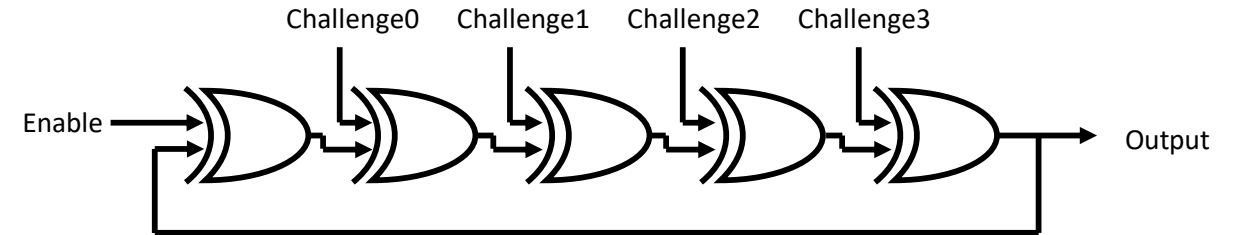
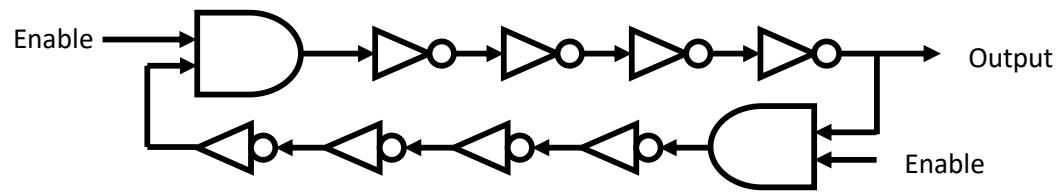
(C)

Low degradation

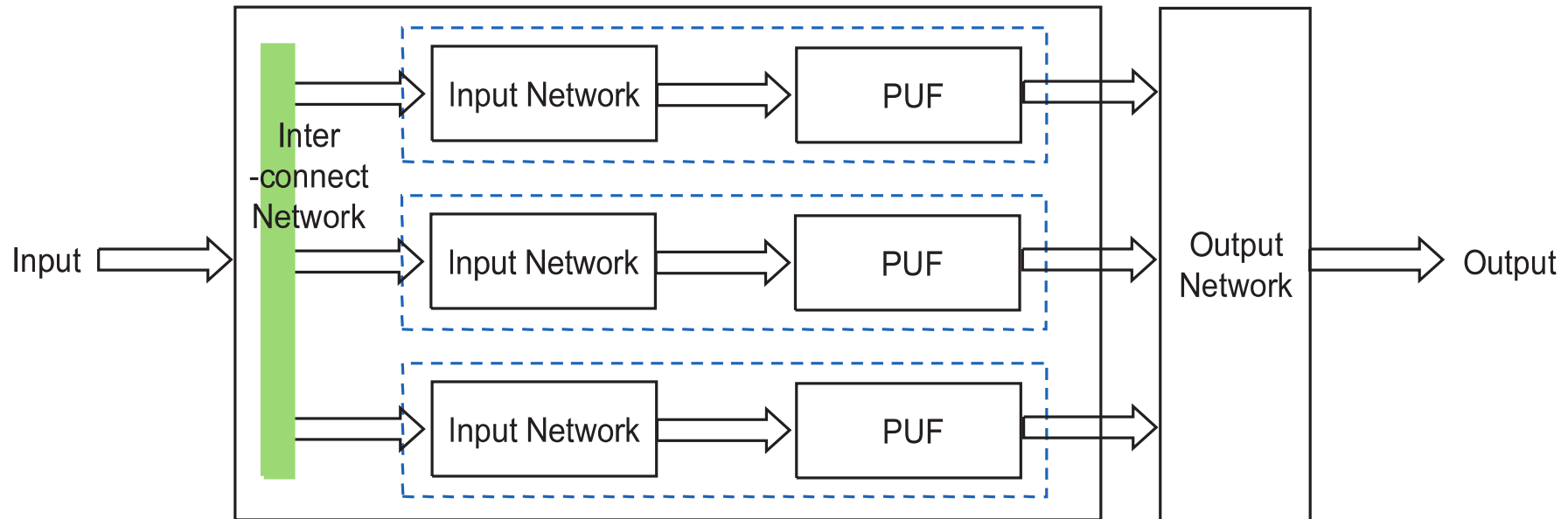
Aging-Resistant RO PUF



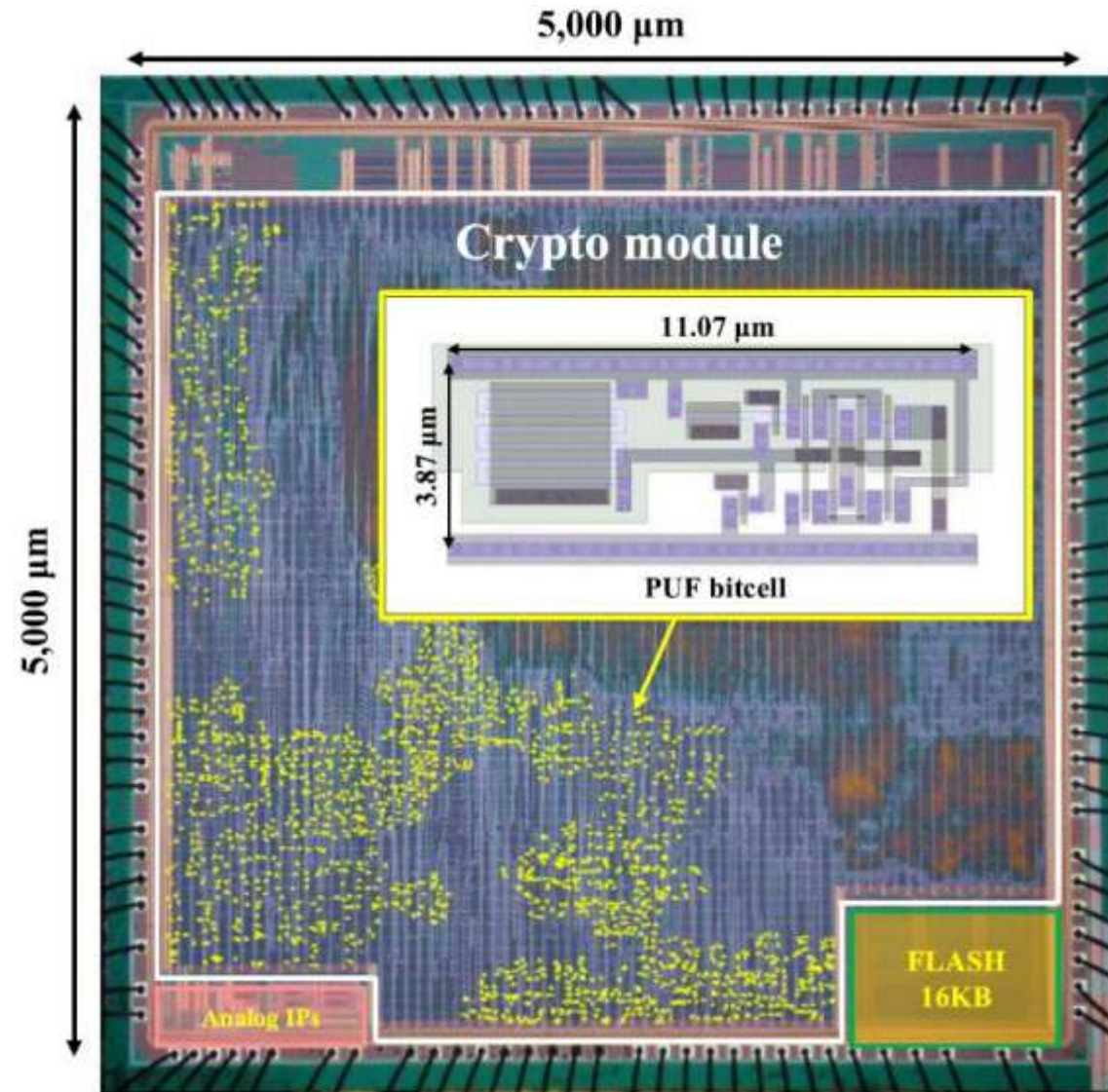
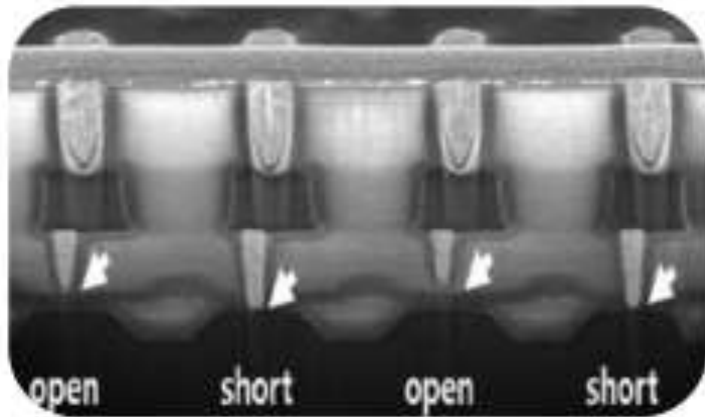
PUF Derivatives



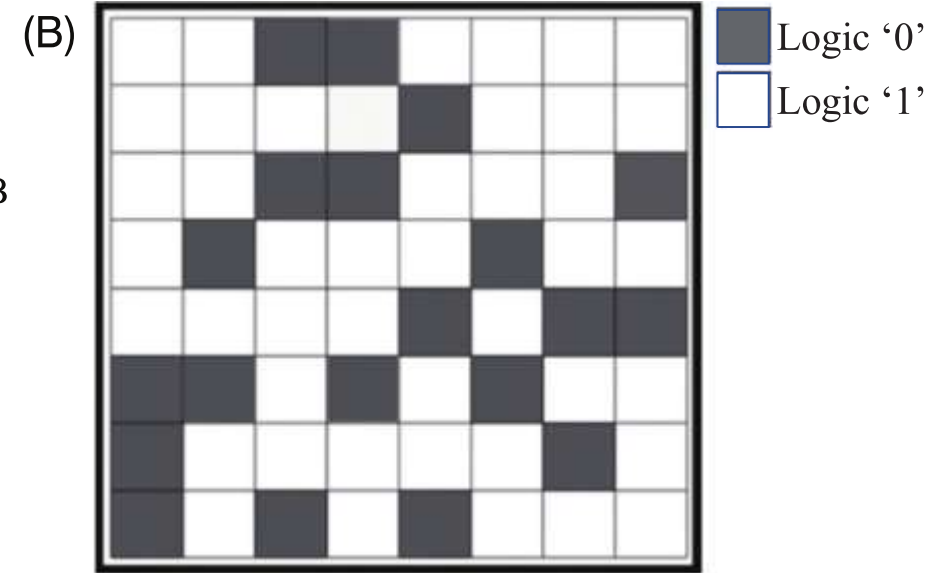
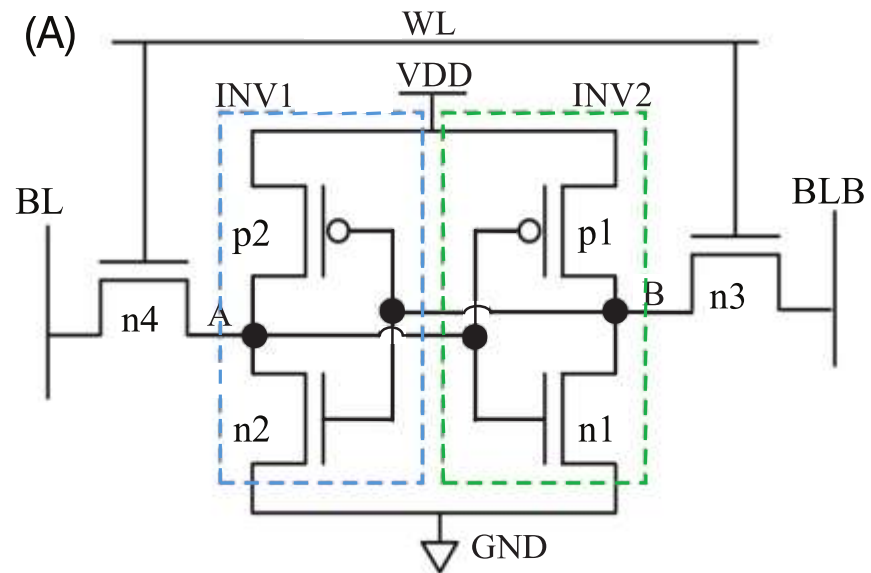
Lightweight PUF



VIA PUF



SRAM PUF

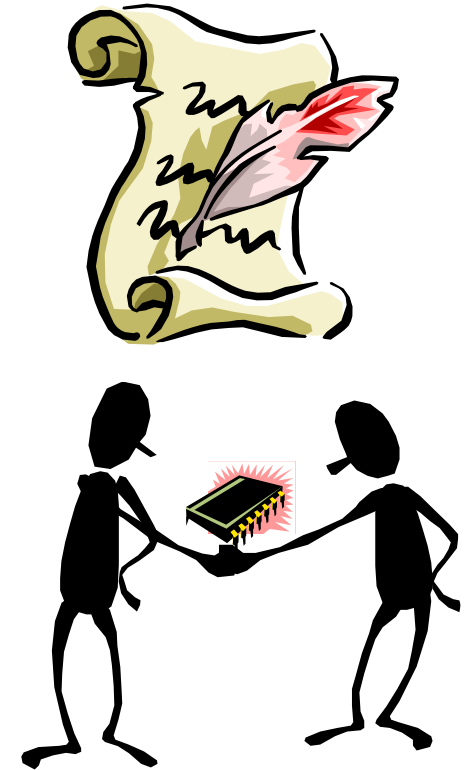


Emerging NVM Properties for PUF

Device	Manufacturing Variability Sources	Exploitable Features	Reliability Factors
PCM	• Geometric Variation (GST layer thickness and Bottom Electrode Contact Diameter) • Cell Resistance (R_{Cell}) • Write/Read Strength (required min. reset current) • Write current magnitude (I_{Write})	• Cell Variability - Stochastic Resistance Variation per cell • Programming sensitivity - variation in resistance level due to the nature of applied pulses	• Power supply Noise and Temperature Variation • Resistance Drift • Poor Endurance and Retention
Memristor & RRAM	• Variability in dope and undoped region length • Device thickness and Cross-sectional area • Stochasticity in doped and undoped resistance.	• Stochastic switching mechanism and intrinsic variability of devices. • Resistance variability due to applied voltage pulse duration	• Power supply Noise and Temperature Variation • Moderate Endurance and Retention • Read Disturbance
MRAM & STTRAM	• Geometric variation in free layer volume • Spin-torque Switching • Threshold voltage • Thermal Stability • Critical Switching Current	• Read Current variation • Meta-stability of free layer magnetization. • Variation in thermal energy • Back-hopping	• Temperature • External EM field • High Endurance and Retention

Applications

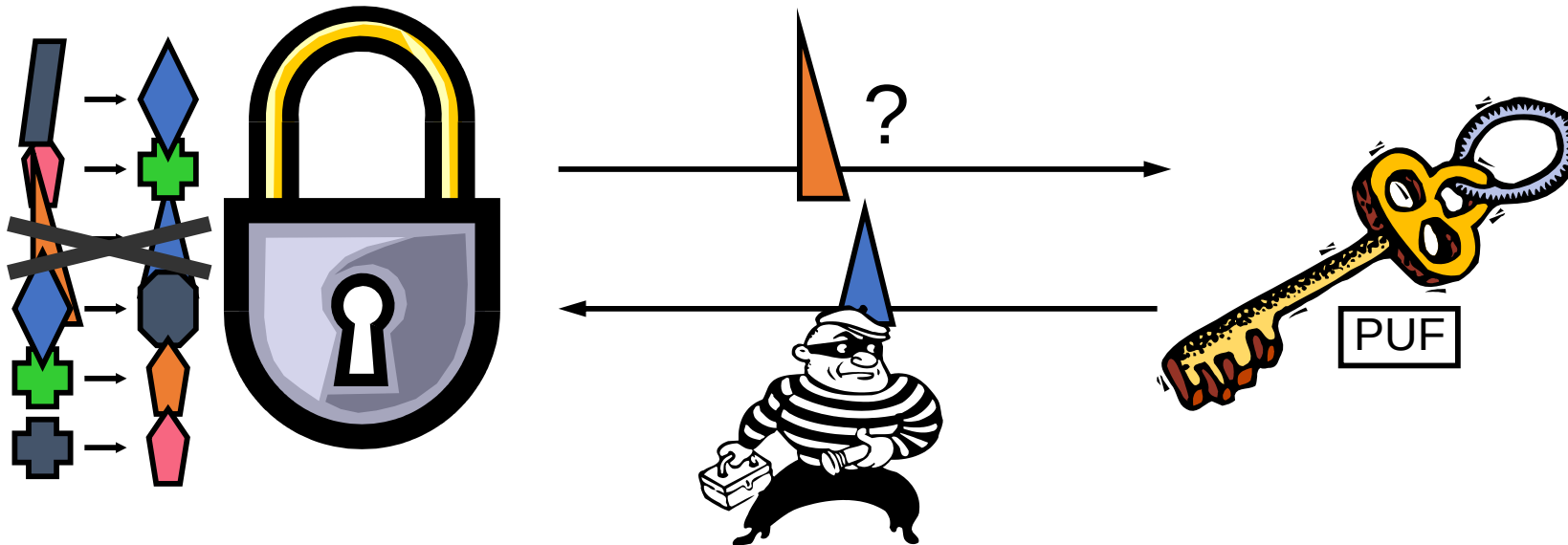
- Anonymous Computation
 - Alice wants to run computations on Bob's computer, and wants to make sure that she is getting correct results. A certificate is returned with her results to show that they were correctly executed.
- Software Licensing
 - Alice wants to sell Bob a program which will only run on Bob's chip (identified by a PUF). The program is copy-protected so it will not run on any other chip.



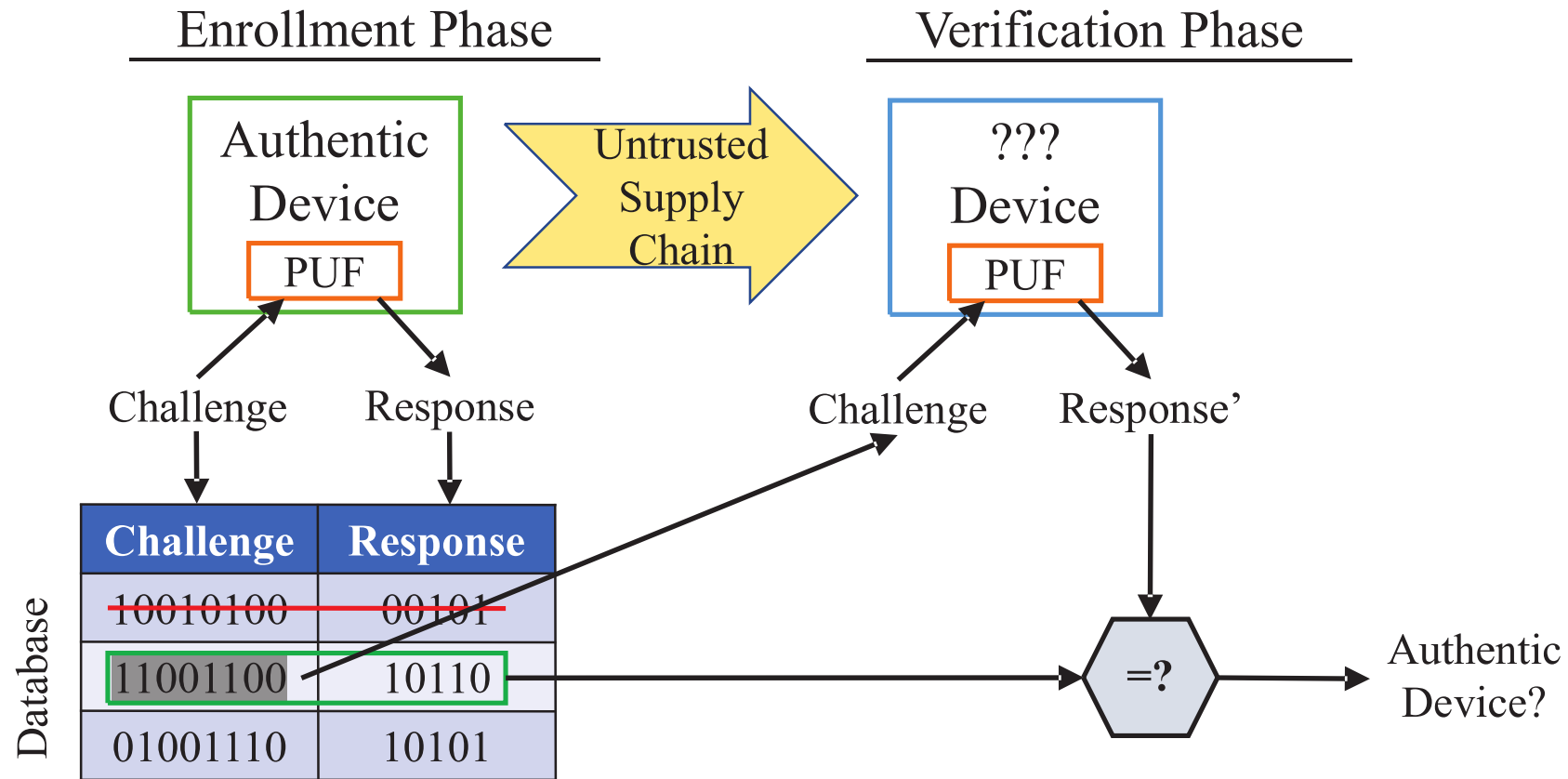
We can enable the above applications by trusting only a single-chip processor that contains a silicon PUF.

Using a PUF as an Unclonable Key

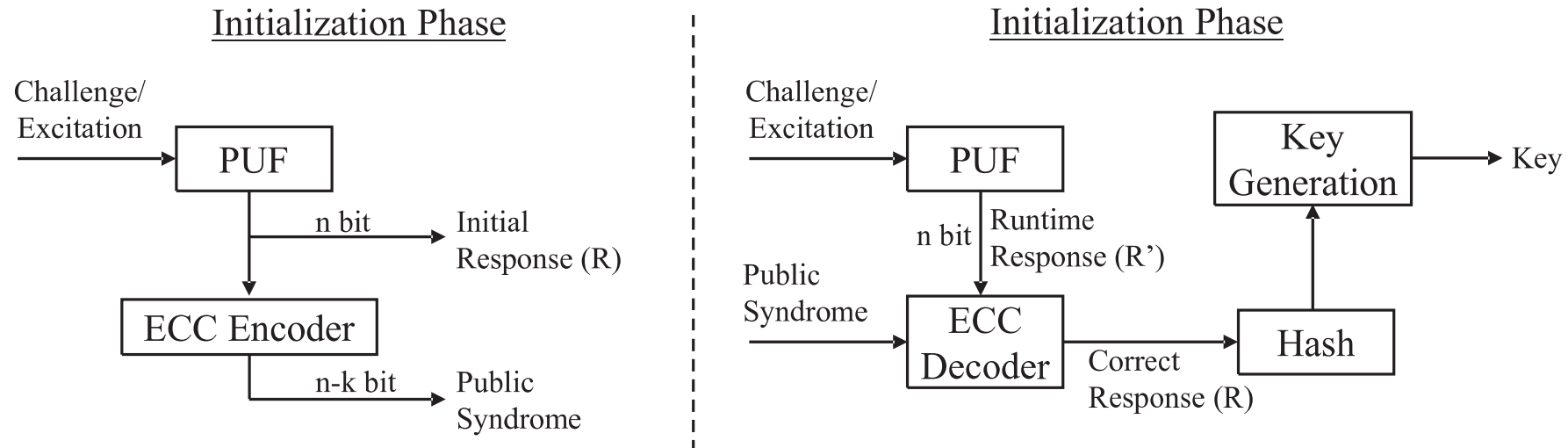
- A silicon PUF can be used as an unclonable key.
 - The lock has a database of challenge-response pairs.
 - To open the lock, the key has to show that it knows the response to one or more challenges.



Strong PUF-based Authentication



Key Generation with PUFs



Random Numbers in Use

Cryptography

Mobile communications, e-banking, online payments, ATMs, wireless keys, cybersecurity, and distributed power grid security all depend on true randomness.

Scientific Computing

Monte Carlo calculations, numerical simulations, statistical research, and randomized algorithms require unpredictable numbers for accurate results.

Gaming & Lotteries

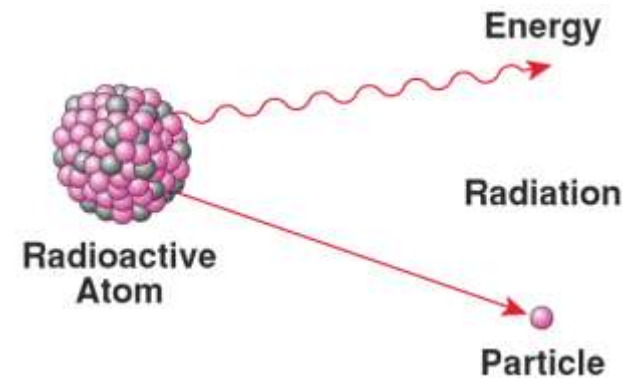
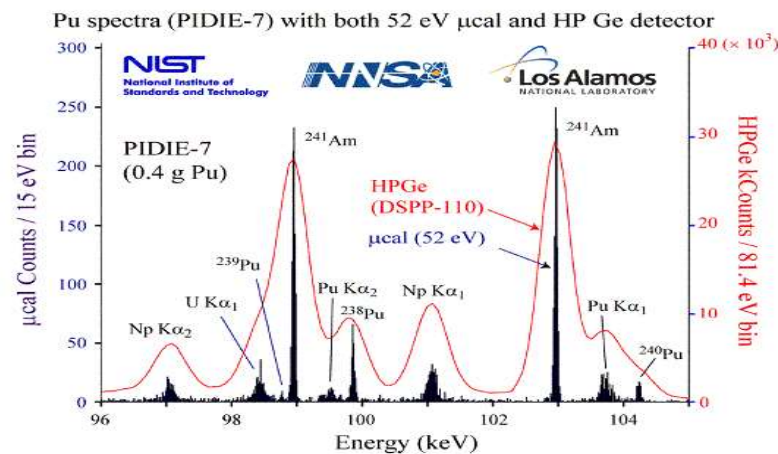
Fair gambling and lottery systems need certified random number generators. Online gambling alone involves an estimated \$6 billion annually in the USA.

Pseudo-random Number Generator

- Pseudo-random number generator:
 - A polynomial-time computable function $f(x)$ that expands a short random string x into a long string $f(x)$ that appears random
- Not truly random in that:
 - Deterministic algorithm
 - Dependent on initial values (seed)
- Objectives
 - Fast
 - Secure

Sources

- The only truly random number sources are those related to physical phenomena such as the rate of radioactive decay of an element or the thermal noise of a semiconductor.
- Randomness is bound to natural phenomena. It is impossible to algorithmically generate truly random numbers.



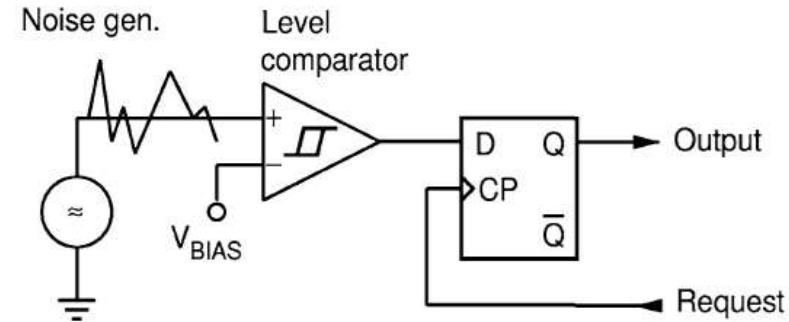
Good TRNG Design

- Entropy Source:
 - Randomness present in physical processes such as thermal and shot noise in circuits, brownian motion, or nuclear decay.
- Harvesting Mechanism:
 - The mechanism that does not disturb the physical process but collects as much entropy as possible.
- Post-Processing (optional):
 - Applied to mask imperfections in entropy sources or harvesting mechanism or to provide tolerance in the presence of environmental changes and tampering.

Set of Requirements

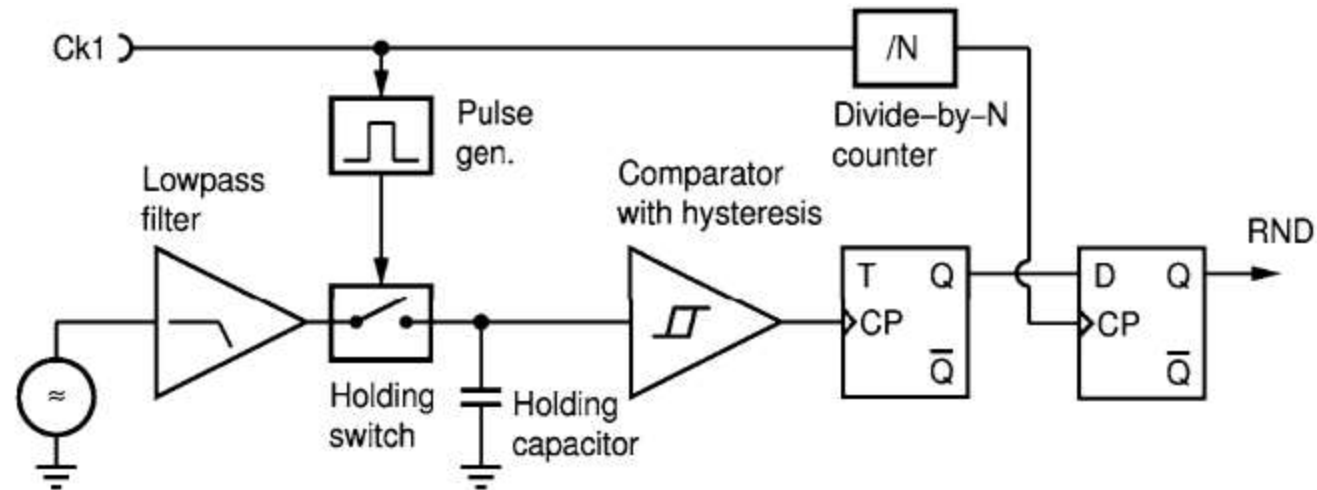
- The design should be purely digital
- The harvesting mechanism should be simple.
 - The unpredictability of the TRNG should not be based on the complexity of the harvesting mechanism, but only on the unpredictability of the entropy source.
- Compact and efficient design (high throughput per area and energy spent).
- The design should be sufficiently simple to allow rigorous analysis.

Noise-Based RNG



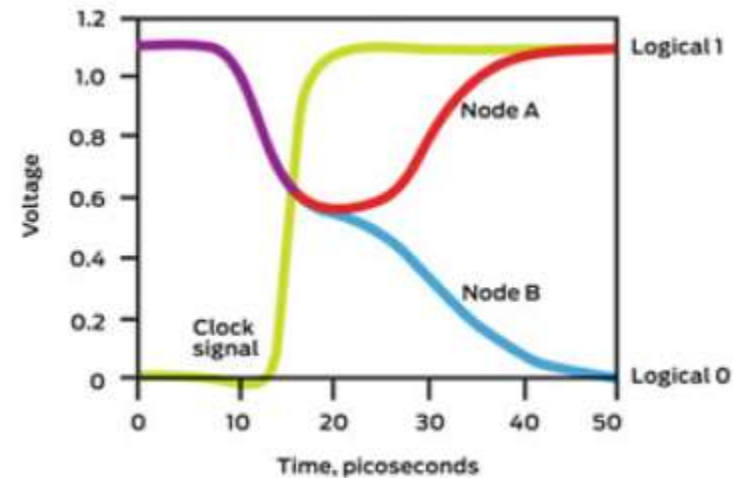
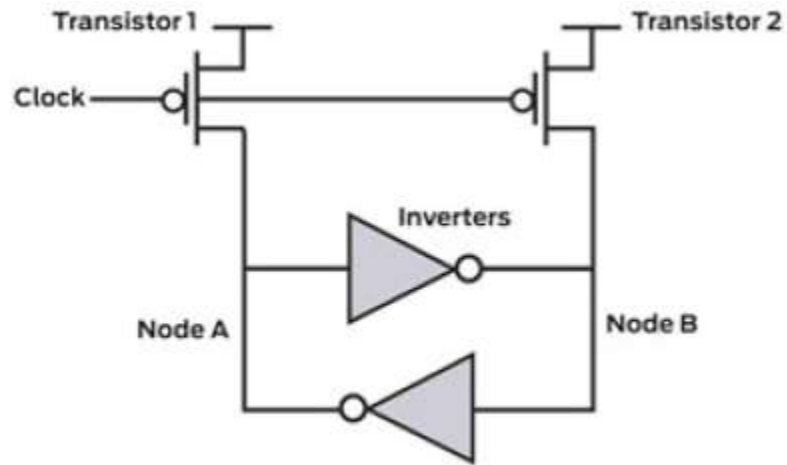
- Noise-based RNGs exploit Johnson's thermal noise in resistors or Zener noise in semiconductor diodes. Johnson's effect creates random voltage from thermal motion of charge carriers at any temperature above absolute zero.
- The fundamental challenge
 - Noise sources cannot be well characterized or controlled during fabrication.
 - Amplification introduces bandwidth limitations and nonlinearity.
 - Multiple nearby RNGs tend to synchronize through electromagnetic interference, dramatically reducing entropy.

Bagini-Bucci Generator



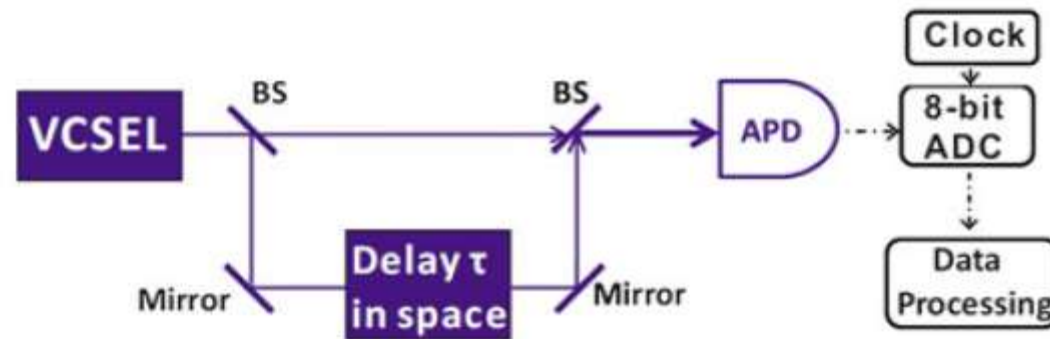
- Uses T flip-flop to divide biased output by 2, spending exactly 50% of time in each state.
- Requires sampling frequency much lower than noise frequency to approach randomness.

Intel's RNG



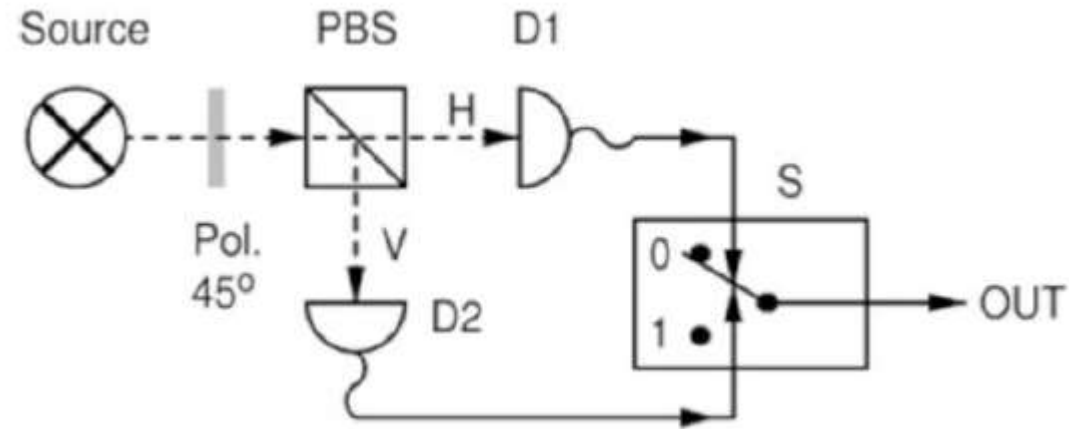
- The circuit has two stable states, but even smallest asymmetries cause high imbalance or complete lockup.

Laser Phase Noise Generators



- Single-mode VCSEL laser with signal and delayed copy brought to interference via Michelson-Morley interferometer.
- Phase jitter detected by high gain-bandwidth photodiode produces electrical noise.

Beam Splitter



- Circularly polarized photon enters polarizing beam splitter (PBS). PBS decomposes polarization, sending horizontal component to one port, vertical to another.
- Circularly polarized photon has equal content of both linear polarizations but cannot split in half—exactly 50% chance to exit either port. Label ports "0" and "1" for theoretically perfect RNG.